## Question #1 - 4 of 54

Question ID: 1674949

Which portfolio had the *best* return to systematic risk ratio?

**A)** Benchmark. ✗

**B)** Portfolio X. ✗

**C)** Portfolio Y. ✓

**Explanation**

The Treynor ratio measures return divided by beta, where beta is a proxy for systematic risk. Portfolio Y has the highest Treynor ratio.

(Module 13.4, LOS 13.p)

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## Question #2 - 4 of 54

Question ID: 1674950

Which statement is *most accurate* regarding Portfolios X and Y?

**A)** Portfolio X has an inferior reward/risk ratio as compared to Portfolio Y. ✗

**B)** Portfolio Y outperformed the benchmark by taking greater risk. ✓

**C)** Portfolio Y more closely resembles the benchmark as compared to Portfolio X. ✗

**Explanation**

Portfolio Y generated higher return than the benchmark, but it produced a lower Sharpe ratio and a lower Sortino ratio, indicating that Portfolio Y experienced greater volatility. Portfolio X has superior reward/risk ratios as measured by Sharpe and Sortino. Portfolio Y has a higher information ratio, indicating that it has less resemblance to the benchmark.

(Module 13.4, LOS 13.p)

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## Question #3 - 4 of 54

Question ID: 1674951

Which portfolio has the *best* return to downside risk ratio?

**A)** Portfolio Y. ✗

**B)** Portfolio X. ✓

**C)** Benchmark.

**Explanation**

The Sortino ratio measures return to downside risk ratio. Portfolio X had the highest Sortino ratio.

(Module 13.4, LOS 13.p)

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## Question #4 - 4 of 54

Question ID: 1674952

An investor seeking to maximize her return and who is indifferent to short-term volatility should *most likely* invest in:

**A)** the benchmark.

**B)** Portfolio Y.

**C)** Portfolio X.

**Explanation**

Portfolio Y generated the highest investment return, but it achieved those results with greater volatility (measured by standard deviation). Given that the investor is not sensitive to volatility, Portfolio Y is most likely the best option.

(Module 13.4, LOS 13.p)

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## Question #5 of 54

Question ID: 1674927

For which of the following is a liability-based benchmark *most likely* appropriate?

**A)** A defined contribution employee retirement fund.

**B)** A defined benefit pension fund.

**C)** A philanthropic endowment with a 5% spending policy.

**Explanation**

A liability-based benchmark is most appropriate for institutions with specific liabilities to pay in the future. A defined benefit pension fund has explicit funding liabilities over a short-, medium-, and long-term time horizon. A defined contribution fund does not have explicit liabilities to manage. An endowment with a 5% spending policy can use a return-based benchmark.

(Module 13.3, LOS 13.i)

## Question #6 of 54

Question ID: 1674902

In global performance evaluation, performance attribution seeks to:

**A)** measure the risk and return of the portfolio. ✗

**B)** identify the sources of difference between portfolio and benchmark return. ✓

**C)** differentiate whether returns come from a manager's luck or skill. ✗

**Explanation**

Performance attribution seeks to identify the sources of difference between portfolio and benchmark return. Note that performance measurement involves the calculation of risk and return, while performance appraisal seeks to identify whether returns are a result of a manager's luck or skill.

(Module 13.1, LOS 13.c)

## Question #7 of 54

Question ID: 1674907

A transactions-based attribution:

**A)** is more complicated than a holdings-based attribution. ✓

**B)** uses the unadjusted portfolio assets from the beginning of a measurement period. ✗

**C)** involves running regressions of broad market indices against portfolio returns to decompose investment performance. ✗

**Explanation**

A returns-based attribution uses regressions of indices against portfolio returns instead of the actual holdings of a portfolio. Holdings-based attribution does not adjust holdings for transactions that occur during a measurement period. Because a transaction-based attribution adjusts the portfolio for subsequent trades, it is more complicated than a holdings-based attribution.

(Module 13.1, LOS 13.d)

## Question #8 of 54

Question ID: 1674904

Which of the following is an example of macro attribution? Quantifying a:

**A)** portfolio manager's decision to bet against emerging-market currencies. ✗

**B)** pension fund's decision to swap a private equity manager for a fixed-income manager because interest rates are low. ✓

**C)** portfolio manager's decision to overweight telecom stocks because interest rates are low. ✗

**Explanation**

Macro attribution analyzes investment decisions at the fund sponsor's level (e.g., a pension fund). Macro attribution quantifies the fund sponsor's decision to deviate from their strategic allocation and the timing of when they made those decisions. Quantifying decisions at the portfolio manager's level is micro attribution.

(Module 13.1, LOS 13.c)

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## Question #9 of 54

Question ID: 1674908

Which of the following is *most correct* regarding an arithmetic attribution?

**A)** The arithmetic attribution is designed to explain the return differential between a portfolio and the S&P 500. ✗

**B)** The arithmetic attribution necessitates adjustments to smooth the subperiod effects over longer periods. ✓

**C)** Arithmetic attribution is better suited than geometric attribution for use with industry professionals. ✗

**Explanation**

The arithmetic attribution should use the most appropriate benchmark, not just the S&P 500. The geometric attribution is more accurate over multiple periods; thus, it is better suited for industry professionals. The arithmetic attribution requires smoothing adjustments over multiple periods.

(Module 13.2, LOS 13.e)

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## Question #10 of 54

Question ID: 1674919

Sarah Cunningham, CFA, uses a top-down analysis process when considering new investments. In performing risk attribution analysis on an absolute basis, the *most appropriate* approach to measuring her performance should focus on:

**A)** attributing tracking error to relative allocation and selection.

**B)** each factor's marginal contribution to total risk and specific risk.

**C)** each security's marginal contribution to total risk.

**Explanation**

When performing absolute risk attribution analysis on managers employing a top-down investment process, we must consider each factor's marginal contribution to total risk and specific risk.

(Module 13.2, LOS 13.g)

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## Question #11 of 54

Question ID: 1674947

Which of the following measures would be the *most appropriate* one to use when comparing the results of two portfolios in which each portfolio contains many stocks from a broad selection of different industries?

**A)** Information ratio.

**B)** Sharpe ratio.

**C)** Treynor measure.

**Explanation**

The equations for the three measures are as follows:

$$\text{Treynor measure} = (R_P - R_F) / \beta_P$$

$$\text{Sharpe ratio} = (R_P - R_F) / \sigma_P$$

$$\text{Information ratio} = (R_P - R_B) / (\sigma_{P-B})$$

Because both portfolios are well diversified, most of their risk comes from systematic risk or beta and is tied to the general level of overall risk in the market. In this case, the best measure to use would be the Treynor measure because this uses beta or systematic risk as the measure of risk. The Sharpe ratio uses standard deviation as the measure of risk in the denominator, and the information ratio is best to use when comparing a portfolio to a benchmark.

(Module 13.4, LOS 13.o)

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## Question #12 of 54

Question ID: 1674943

The Treynor measure is correctly defined as a measure of a fund's:

**A)** return earned compared to its systematic risk.

**B)** excess earned compared to its systematic risk.

**C)** return earned compared to its unsystematic risk.

**Explanation**

The Treynor measure is defined as a fund's excess return (fund's return minus the risk-free rate) divided by its systematic risk (beta).

(Module 13.4, LOS 13.o)

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## Question #13 of 54

Question ID: 1674901

An effective performance attribution process *most likely* includes:

**A)** the portfolio manager's prior decision-making process.

**B)** the active investment decisions taken by the portfolio manager.

**C)** a partial explanation of the portfolio's excess return and risk.

**Explanation**

An effective performance attribution process includes a reflection of 100% of the portfolio's return or risk exposure, the portfolio manager's current decision-making process (not prior decision-making process), the active investment decisions taken by the portfolio manager, and a full explanation of the portfolio's excess return and risk. (Module 13.1, LOS 13.b)

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## Question #14 of 54

Question ID: 1674899

What is the goal of performance appraisal?

**A)** Identification of overall risk and return.

**B)** Interpretation of performance attribution.

**C)** Identification of the sources of differences between portfolio and benchmark risk and return.

**Explanation**

Performance appraisal involves the interpretation of performance attribution. A judgment is made about manager's decisions and skill, in an effort to differentiate between returns attributable to luck and those attributable to skill.

(Module 13.1, LOS 13.a)

## Question #15 of 54

Question ID: 1674906

Holdings-based attribution is *most appropriate* for assessing:

**A)** holdings data based on long time intervals.

**B)** passive index funds.

**C)** portfolios with high turnover.

**Explanation**

Holdings-based attribution does not make any adjustments for portfolio changes. Hence, it is most widely applied when analyzing portfolios with low turnover, such as passive index funds. The accuracy of holdings-based attribution improves when using data with shorter time intervals. (When transactions are made during the measurement period, holdings-based attribution may not reconcile to actual portfolio return.)

(Module 13.1, LOS 13.d)

## Question #16 of 54

Question ID: 1674920

Lisa Roberts is trying to determine whether her equity investment managers have added any value, so she has decided to conduct a macro attribution analysis of their portfolio returns. Roberts has gathered the following results:

| | Fund Weight | Fund Return | Benchmark Weight | Benchmark Return |
|---|---|---|---|---|
| **Growth Manager** | **40%** | 9.75% | **35%** | 8% |
| Small-cap growth | 25% | 12% | 20% | 8% |
| Mid-cap growth | 15% | 6% | 15% | 8% |
| **Value Manager** | **60%** | 6% | **65%** | 3.85% |
| Small-cap value | 40% | 7% | 35% | 2% |
| Mid-cap value | 20% | 4% | 30% | 6% |
| **TOTAL** | **100%** | **7.5%** | **100%** | **5.3%** |

Given the results compiled, the allocation effect of Roberts's underweighting value in her portfolio using the Brinson-Fachler model is *closest* to:

**A)** −5 basis points.

**B)** 14 basis points.

**C)** 7 basis points.

**Explanation**

Using the Brinson-Fachler model allocation effect = (0.6 − 0.65) × (3.85% − 5.3%) = 0.0725% = 7.25 bp.

(Module 13.2, LOS 13.h)

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## Question #17 of 54

Question ID: 1674941

Which statement regarding the Sharpe ratio and Sortino ratio is *most accurate*?

**A)** In both the Sharpe ratio and Sortino ratio, a higher calculated ratio is better.

**B)** The Sortino ratio only considers upside risk, while the Sharpe ratio considers both upside and downside risk.

**C)** In both the Sharpe ratio and Sortino ratio, the risk-free rate is used as a return benchmark.

**Explanation**

In both Sharpe and Sortino ratios, higher calculated ratios indicate superior risk/reward measures. The Sortino ratio only considers downside risk and uses a target return instead of the risk-free-rate.

(Module 13.4, LOS 13.n)

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## Question #18 of 54

Question ID: 1674932

Which of the following is the *most appropriate* method of calculating the manager's active return? The manager's active return is the:

**A)** market return minus the benchmark return.

**B)** portfolio return minus the benchmark return.

**C)** portfolio return minus the market return.

**Explanation**

The manager's active return is the portfolio return minus the benchmark return, where the benchmark is appropriate to the manager's style.

(Module 13.3, LOS 13.k)

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## Question #19 of 54

Question ID: 1674914

For the purposes of risk attribution, an analysis of a top-down investor would *most likely* include:

**A)** tracking error resulting from overweighting the technology sector. ✓

**B)** tracking error resulting from underweighting several stocks relative to the benchmark. ✗

**C)** a security's marginal contribution to standard deviation. ✗

**Explanation**

Risk attribution analysis of a top-down manager would focus on differentials from allocations to sectors and asset classes. An analysis of a bottom-up manager would focus on tracking error and incremental risk from the selection of individual securities as would a security's marginal contribution to standard deviation.

(Module 13.2, LOS 13.g)

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## Question #20 of 54

Question ID: 1674928

Which of the following is not regarded to be an essential characteristic of a valid benchmark?

**A)** Specified in advance. ✗

**B)** Appropriate to the manager’s investment approach and style. ✗

**C)** Reflective of past investment opinion. ✓

**Explanation**

The benchmark has seven characteristics. All of the listed options are included with the exception of "reflective of past investment opinion"—it should be reflective of *current* investment opinion, and the manager should have *current* knowledge and expertise of the securities in the benchmark.

(Module 13.3, LOS 13.j)

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## Question #21 of 54

Question ID: 1674942

Analysis of a portfolio's drawdown is useful for evaluating:

**A)** how long it will take to recover after a period of losses. ✓

**B)** a portfolio's downside capture relative to an index. ✗

**C)** how much risk a portfolio manager is taking relative to returns. ✗

**Explanation**

Drawdown analysis can be used to evaluate the maximum drawdown of a portfolio (peak to trough) and the total time required to fully recover. Drawdown analysis generally does not take returns relative to risk or returns relative to an index into consideration.

(Module 13.4, LOS 13.n)

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## Question #22 of 54

Question ID: 1674926

A liability-based benchmark is *most likely* to include:

**A)** nominal bonds only. ✗

**B)** both nominal and inflation-adjusted bonds. ✓

**C)** inflation-adjusted bonds only. ✗

**Explanation**

Investment managers having a specific liability to fund in the future use liability-based benchmarks. Assets included in these benchmarks represent typical investments by these managers, and include nominal bonds, inflation-adjusted bonds, and lower-volatility stocks.

(Module 13.3, LOS 13.i)

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## Question #23 of 54

Question ID: 1674933

XYZ Asset Management operates a Pan-European corporate bond fund. Last year, the bond fund generated a 5.1% return. In the same year, the bond fund's benchmark generated a 6.0% return; however, the broad index of European bonds generated a 4.5% return. XYZ's bond fund return due to active management is *closest* to:

**A)** −0.9%. ✓

**B)** 0.6%.

**C)** 1.5%.

**Explanation**

Active return = the difference between the manager's overall return and the style benchmark return. XYZ's bond fund returned 5.1%, and its benchmark returned 6.0%. Therefore, the active return = (5.1% − 6.0%) = −0.9%.

(Module 13.3, LOS 13.k)

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## Question #24 of 54

Question ID: 1674929

Fund Sponsors often use the median account in a particular universe of account returns as an appropriate benchmark. This form of benchmark has numerous drawbacks. Which of the following is not a drawback that would be associated with using the median account as a benchmark?

**A)** It is not measurable as its value cannot be determined on a reasonably frequent basis.

**B)** As the median manager is unknown, the measure is ambiguous.

**C)** It is virtually impossible to identify the median manager in advance.

**Explanation**

There are seven properties of a valid benchmark. Regarding the median account approach, its value is measurable. This is probably the only criteria that the median manager approach satisfies. The other statements are true of the median account.

(Module 13.3, LOS 13.j)

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## Question #25 of 54

Question ID: 1674900

Which component of performance evaluation determines whether the performance was affected primarily by skill or luck?

**A)** Performance appraisal.

**B)** Performance measurement.

**C)** Performance attribution.

**Explanation**

Performance appraisal determines whether the performance was affected primarily by investment decisions, by the overall market, or by chance. Performance measurement refers to the calculation of return and risk over specified periods. Performance attribution expands upon the performance measurement and explains how the return was achieved given the risk taken. (Module 13.1, LOS 13.a)

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## Question #26 of 54

Question ID: 1674931

A retired investor needs to earn at least $100,000 per year in investment income to fund living expenses. Which type of benchmark is *best* suited for this individual?

**A)** Returns-based benchmark. ✗

**B)** Absolute benchmark. ✓

**C)** Broad market index. ✗

**Explanation**

An absolute benchmark is best for investors who aim to exceed a minimum target return. In this example, the investor needs to earn a minimum absolute return of $100,000 to fund expenses.

(Module 13.3, LOS 13.j)

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## Question #27 - 30 of 54

Question ID: 1674922

Based upon the data in Exhibit 1, which of the following statements is *least accurate*?

**A)** The benchmark is a broad-based index with average market risk. ✗

**B)** The manager's large-cap tilt contributed negatively to returns. ✓

**C)** The benchmark has a large-cap bias. ✗

**Explanation**

The benchmark has a negative exposure to small caps, meaning a positive exposure to large caps.

The manager's tilt away from small caps to large caps contributed positively to returns, as small-cap returns were negative during the period (–4.25%).

The benchmark has a large-cap bias without any growth or value tilt. This indicates the benchmark has broad market exposures and average market risk.

(Module 13.2, LOS 13.h)

## Question #28 - 30 of 54

Question ID: 1674923

Which of the following is *closest* to the return from security selection?

**A)** –0.35%.

**B)** +0.35%.

**C)** –0.45%.

**Explanation**

| Factor | Portfolio | Benchmark | Difference | Factor Return | Absolute | Proportion of Total |
|---|---|---|---|---|---|---|
| RMRF | 0.97 | 1.00 | –0.03 | 6.45% | –0.1935% | |
| SMB | –1.08 | –1.00 | –0.08 | –4.25% | 0.3400% | |
| HML | 0.35 | 0.00 | 0.35 | 5.50% | 1.9250% | 141.65% |
| WML | –0.01 | 0.02 | -0.03 | 8.75% | –0.2625% | |
| Factor tilts return | | | | | 1.8090% | |
| Security selection return | | | | | –0.4500% | |
| Active return | | | | | 1.3590% | |

Absolute return = difference × factor return (e.g., RMRF = –0.03 × 6.45% = –0.1935%)

Active return = factor tilts return + security selection return

1.3590% = 1.8090% + security selection return

Security selection return = –0.45%.

(Module 13.2, LOS 13.h)

## Question #29 - 30 of 54

Question ID: 1674924

Which of the following is the proportion of total active return contributed by the value factor?

**A)** 141.65%.

**B)** 106.41%.

**C)** 1.925%.

**Explanation**

| Factor | Portfolio | Benchmark | Difference | Factor Return | Absolute | Proportion of Total |
|---|---|---|---|---|---|---|
| RMRF | 0.97 | 1.00 | –0.03 | 6.45% | –0.1935% | |
| SMB | –1.08 | –1.00 | –0.08 | –4.25% | 0.3400% | |
| HML | 0.35 | 0.00 | 0.35 | 5.50% | 1.9250% | 141.65% |
| WML | –0.01 | 0.02 | –0.03 | 8.75% | –0.2625% | |
| Factor tilts return | | | | | 1.8090% | |
| Security selection return | | | | | –0.4500% | |
| Active return | | | | | 1.3590% | |

Value factor (HML, book-to-price) contribution = 1.9250% / 1.3590% = 141.65%.

(Module 13.2, LOS 13.h)

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## Question #30 - 30 of 54

Question ID: 1674925

Which of the following would have had a greater positive impact on overall return compared to the manager's actual return?

**A)** Increasing exposure to the market factor to 1.06. ✓

**B)** Increasing the stock selection return by +0.55%. ✗

**C)** Increasing exposure to momentum to 0.04. ✗

**Explanation**

Increasing exposure to the momentum factor: (0.04 –(–0.01) × 8.75% = 0.4375%

Increasing the stock selection by +0.55% = 0.550%

Increasing exposure to the market factor: (1.06 – 0.97) × 6.45% = **0.5805%**.

Of the three answer choices provided, increasing the exposure to the market factor to 1.06 had the largest improvement on return.

(Module 13.2, LOS 13.h)

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## Question #31 of 54

Question ID: 1674905

Regression analysis is *most likely* to be used to analyze portfolio returns when performing:

**A)** returns-based attribution. ✓

**B)** transactions-based attribution. ✗

**C)** holdings-based attribution. ✗

**Explanation**

Regression analysis is used to determine how portfolio returns compare with index returns in returns-based attribution. Holdings-based attribution considers the holdings over time to evaluate the decisions that contributed to returns, while transactions-based attribution uses both holdings and transactions to explain performance.

(Module 13.1, LOS 13.d)

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## Question #32 of 54

Question ID: 1674910

Jonathan Kennedy is performing attribution analysis of a three-sector domestic equity portfolio. The following table shows data from the portfolio along with benchmark data.

| Sector/Industry | Weight in Portfolio | Weight in Benchmark | Return From Portfolio | Return From Benchmark |
|---|---|---|---|---|
| Biotech | 30% | 25% | 5% | 3% |
| Utilities | 40% | 35% | 10% | 1% |
| Media | 30% | 40% | -5% | -4% |
| **Total** | 100% | 100% | -4% | -0.5% |

Using the data in the table, the total selection effect is *closest* to:

**A)** 3.75%. ✗

**B)** 4.25%. ✗

**C)** 3.25%. ✓

**Explanation**

The selection effect calculation is the same in both the Brinson-Fachler and Brinson-Hood-Beebower (BHB) models. The total selection effect for the portfolio is determined by summing all of the contributions to the selection, such that $S_i = W_i \times (R_i - B_i)$. From the table, we can see that the individual sector selection effects are:

- Biotech: $S_i = W_i \times (R_i - B_i) = 25\% \times (5\% - 3\%) = 0.5\%$
- Utilities: $S_i = Wi \times (R_i - B_i) = 35\% \times (10\% - 1\%) = 3.15\%$
- Media: $S_i = W_i \times (R_i - B_i) = 40\% \times (-5\% - (-4\%)) = -0.4\%$
- TOTAL = **3.25%**

(Module 13.2, LOS 13.e)

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## Question #33 of 54

Question ID: 1674939

One reason it can be difficult to benchmark hedge funds is because hedge funds:

**A)** frequently invest in illiquid assets. ✓

**B)** generally perform in line with their peer universe. ✗

**C)** generally underperform the broad market indices. ✗

**Explanation**

Many hedge funds invest in illiquid assets; illiquid assets are difficult to corroborate the value of underlying securities because there is not a traded market. Pricing illiquid securities generally involves a third-party appraisal, which may occur infrequently and is difficult to track over time. Hedge funds often vary quite a great deal from peers, even within the same markets. Hedge fund underperformance relative to the broad market indices is not relevant to their benchmark.

(Module 13.3, LOS 13.m)

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## Question #34 - 37 of 54

Question ID: 1674954

Which of the following statements regarding the use of the median manager's results as a benchmark for evaluating performance of a portfolio is *most accurate*?

**A)** It is not possible to identify the median manager. ✗

**B)** The weights of individual securities required to replicate the median manager cannot be known. ✗

**C)** The performance of the median manager is subject to upward bias. ✓

**Explanation**

At first glance, all three answer choices could seem accurate. It is true that the median manager is generally a poor choice as a benchmark because one cannot know who that manager is or invest with them *in advance*. However, *after the fact*, one does know who the median manager is, and one can likely find out the asset holdings and weights of that manager. So, two of the statements are partially inaccurate.

The most accurate statement is the upward bias of reported median manager results. Manager universes are subject to many biases, including self-reporting and survivor bias. Better managers tend to report, and failing manager results are often removed from the historical record. Both of those effects would bias upward the results of the reported median manager.

(Module 13.3, LOS 13.j)

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## Question #35 - 37 of 54

Question ID: 1674955

The Treynor ratio for the Knight Fund is *closest* to:

**A)** 7.56.

**B)** 0.90.

**C)** 4.22.

**Explanation**

$$T_P = \frac{\overline{R}_A - \overline{R}_F}{\beta_A}$$

.

$$T_P = \frac{6.8 - 3.0}{0.9} = 4.22$$

(Module 13.4, LOS 13.n)

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## Question #36 - 37 of 54

Question ID: 1674956

The Sharpe ratio for the Knight Fund is *closest* to:

**A)** 0.90.

**B)** 0.21.

**C)** 0.12.

**Explanation**

$$S_P = \frac{\overline{R}_A - \overline{R}_F}{\sigma_A}$$

.

$$S_P = \frac{6.8 - 3.0}{32.6} = 0.12$$

(Module 13.4, LOS 13.n)

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## Question #37 - 37 of 54

Question ID: 1674957

Which performance measure would Nero choose if Nero wanted to consider the impact of both market-related variability of returns and the variability of active management?

**A)** Treynor ratio.

**B)** Information ratio.

**C)** Sharpe ratio.

**Explanation**

The request covers total variability of return, and that is measured by standard deviation. Standard deviation includes the impact of market-related (beta) and active management–related (alpha) variability of return. The Sharpe ratio uses standard deviation in the denominator.

The Treynor ratio considers only beta risk, while the information ratio considers only alpha-related risk.

(Module 13.4, LOS 13.n)

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## Question #38 of 54

Question ID: 1674909

The *most common* approach for calculating excess returns for individual investor portfolios involves calculating:

**A)** arithmetic attribution results.

**B)** total return attribution results.

**C)** geometric attribution results.

**Explanation**

In general, arithmetic attribution is most appropriate for nontechnical investors such as individuals because it is simpler to calculate than geometric attribution results. Arithmetic attribution approaches are designed to convey the arithmetic difference between the portfolio return and its benchmark return.

(Module 13.2, LOS 13.e)

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## Question #39 of 54

Question ID: 1674911

Ha-Yoon Kim is a stock analyst with Yeongli Securities Co. She uses the Carhart factor model to perform attribution analysis. Output from the model indicates the benchmark that Kim uses has the following factor sensitivities:

| Factor | Benchmark |
|---|---|
| RMRF | +1.00 |
| SMB | −1.00 |
| HML | 0.05 |
| WML | −0.95 |

Assuming that she is using a benchmark that most closely resembles her portfolio attributes, we can conclude that Kim is *most likely* a:

**A)** large-cap investor.

**B)** small-cap investor.

**C)** momentum investor.

**Explanation**

The negative coefficient of the benchmark's sensitivities to a market-capitalization factor (SMB) and a momentum factor (WML) indicate that the portfolio has a large-cap tilt and does not seek momentum strategies.

(Module 13.2, LOS 13.e)

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## Question #40 of 54

Question ID: 1674946

Which of the following statements regarding the Sharpe ratio is *most accurate*?

**A)** Beta is not a component of the Sharpe ratio.

**B)** The measure of risk used in the denominator of the Sharpe ratio is standard deviation, also known as unsystematic risk. ✗

**C)** The denominator of the Sharpe ratio is standard deviation, which comprises partly systematic risk called beta. ✓

**Explanation**

The equation for the Sharpe ratio = $(R_P - R_F) / \sigma_P$.

The Sharpe ratio contains standard deviation in the denominator of the equation, which is total risk, and comprises both systematic risk called beta and unsystematic risk. Thus, the Sharpe ratio does contain a component of beta.

(Module 13.4, LOS 13.o)

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## Question #41 - 43 of 54

Question ID: 1674916

Based on the Brinson-Fachler model, the allocation effect of hiring the domestic equities portfolio manager is *closest* to:

**A)** −0.15%. ✗

**B)** 0.15%. ✓

**C)** 0.11%. ✗

**Explanation**

The allocation effect using the Brinson-Fachler model is the difference between the portfolio and the benchmark weights multiplied by the difference between the sector return for the benchmark and the overall benchmark return. Allocation effect = (0.50 − 0.75) × (6.20% − 6.78%) = 0.15%.

(Module 13.2, LOS 13.g)

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## Question #42 - 43 of 54

Question ID: 1674917

The selection and interaction effect of hiring the international equities portfolio manager is *closest* to:

**A)** 0.00%. ✗

**B)** −0.19%. ✓

**C)** 0.09%. ✗

**Explanation**

Selection + interaction effect = the benchmark weight multiplied by the difference in the portfolio and benchmark return plus the difference in the portfolio and benchmark weights multiplied by the difference between the portfolio and benchmark sector returns. Selection + interaction effect = [0.25 × (8.14% − 8.51%)] + [(0.50 − 0.25) × (8.14% − 8.51%)] = −0.19%.

(Module 13.2, LOS 13.g)

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## Question #43 - 43 of 54

Question ID: 1674918

Which statement *most accurately* describes the impact from asset allocation and manager selection in the previous table?

**A)** Value was added with manager selection, and value was added with asset allocation. ✗

**B)** Value was detracted by manager selection, but value was added with asset allocation. ✓

**C)** Value was added with manager selection, but value was detracted with asset allocation. ✗

**Explanation**

The fund selection underperformed the benchmarks; however, selecting the asset class with stronger performance (international equities) added to overall performance and enabled the organization to beat its overall benchmark.

(Module 13.2, LOS 13.g)

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## Question #44 of 54

Question ID: 1674945

Peter Michaels, CFA, works at Composite Consulting, and is in charge of evaluating the performance of various portfolio managers. His main tasks are to measure and evaluate the sources of return that can be attributed to manager performance. Michaels understands the importance of incorporating risk into his analyses, but he realizes the limitations associated with some performance measurement techniques in accomplishing that particular objective. Michaels begins the evaluation of numerous managers by examining return information from both the portfolio being evaluated and its designated benchmark.

Michaels has the following return information for the AM Growth Fund:

| | AM Growth Fund | S&P 500 |
|---|---|---|
| Return | 14% | 12% |
| Standard deviation | 25% | 18% |
| Beta | 1.15 | 1.00 |

If the AM Growth Fund is considered to be well diversified, which measure would be *most appropriate* in evaluating its risk/return performance?

**A)** The Treynor measure. ✓

**B)** The $M^2$ measure. ✗

**C)** The Sharpe ratio. ✗

**Explanation**

If the AM Growth Fund is well diversified, the appropriate risk measure would be beta, or the systematic risk component of total risk. Therefore, the Treynor measure would be appropriate in this case.

(Module 13.4, LOS 13.o)

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## Question #45 of 54

Question ID: 1674944

Which of the following statements about risk/return investment manager performance measures is *least accurate*?

**A)** The Treynor measure includes company-specific risk as part of its performance measurement. ✓

**B)** The Sharpe measure includes company-specific risk as part of its performance measurement. ✗

**C)** When measuring the performance of an equity fund, if the Sharpe ratio is 0.55, and the Treynor measure is 0.47, the difference is attributable to unsystematic (company-specific) risk.

**Explanation**

The Treynor measure does not include company-specific risk—it uses beta in the denominator, which only measures systematic risk. Note that the Sharpe measure uses standard deviation in its denominator, which is a measure of total risk.

(Module 13.4, LOS 13.o)

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## Question #46 - 49 of 54

Question ID: 1674935

Rank the true active returns from the three portfolios from *highest* to *lowest*.

**A)** Gamma, Theta, Delta.

**B)** Gamma, Delta, Theta.

**C)** Theta, Delta, Gamma.

**Explanation**

**Delta has a U.S. small-cap style**, so the closest benchmark is the S&P SmallCap 600:

> True alpha = portfolio return – style specific benchmark return (a.k.a. manager's normal portfolio) = 9.20% – 8.30% = **0.90%**

**Gamma has a U.S. momentum style**, so the closest benchmark is the S&P 500 Momentum:

> True alpha = –5.80% – (–7.10%) = **1.30%**

**Theta has a U.S. large-cap growth style**, so the closest benchmark is the S&P 500 Growth:

> True alpha = 6.70% – 9.50% = **–2.80%**

Ranking the three funds by their true active returns (highest to lowest):

> **Gamma, Delta, Theta.**

(Module 13.3, LOS 13.k)

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## Question #47 - 49 of 54

Question ID: 1674936

**Calculate** the style return for the Delta portfolio.

**A)** 0.90%.

**B)** 0.40%.

**C)** –0.50%.

**Explanation**

Style return = manager's normal portfolio return (style matched benchmark) – broad market return

Delta style return = S&P SmallCap 600 – S&P 500

= 8.30% – 8.80%

= **–0.50%**

U.S. small caps **underperformed the broad market by 0.50%** over the period.

(Module 13.3, LOS 13.k)

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## Question #48 - 49 of 54

Question ID: 1674937

Which of the portfolios has misfit active return of 0.70%?

**A)** Delta.

**B)** Gamma.

**C)** Theta.

**Explanation**

**Misfit active return** is another term for the **style return** and is defined as follows:

> Misfit return = manager's normal portfolio return (style matched benchmark) – broad market return
>
> **Delta misfit active return** = S&P SmallCap 600 – S&P 500
>
> = 8.30% – 8.80% = **–0.50%**
>
> **Gamma misfit active return** = S&P 500 Momentum – S&P 500
>
> = –7.10% – 8.80% = **–15.9%**
>
> **Theta misfit active return** = S&P large-cap growth – S&P 500
>
> = 9.50% – 8.80% = **0.70%**

(Module 13.3, LOS 13.k)

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## Question #49 - 49 of 54

Question ID: 1674938

Which manager's normal portfolio return is the *highest*?

**A)** Gamma.

**B)** Delta.

**C)** Theta.

**Explanation**

A manager's normal portfolio is an appropriate style match passive benchmark. Of the three style benchmarks, Theta has the highest return (9.5%).

(Module 13.3, LOS 13.k)

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## Question #50 of 54

Question ID: 1674940

In the Sortino ratio, the excess return is divided by the:

**A)** standard deviation using only the returns below a minimum level.

**B)** maximum drawdown.

**C)** standard deviation.

**Explanation**

The Sortino ratio examines the downside risk of returns. It is calculated as the portfolio return minus the minimum acceptable return (MAR) divided by a standard deviation that only uses returns below the MAR. It is similar to the target semivariance. Both remaining responses refer to other measures of risk-adjusted performance. The Sharpe ratio divides the excess return above the risk-free rate by the standard deviation. The RoMAD (return over maximum drawdown) is the average portfolio return divided by the maximum drawdown. *Drawdown* refers to the percentage difference between the highest and lowest portfolio values during a period.

(Module 13.4, LOS 13.n)

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## Question #51 of 54

Question ID: 1674913

The following table shows attribution results from a fixed-income portfolio:

| Duration Bucket | Sector | Duration Effect | Curve Effect | Total Interest Rate Allocation Effect | Sector Allocation | Bond Selection | Total |
|---|---|---|---|---|---|---|---|
| **Short** | Corporate | | | | | | |
| | ABS/MBS | | | | | | |
| | **Total** | −0.25% | +0.45% | +0.20% | +0.75% | +0.15% | +1.10% |
| **Long** | Corporate | | | | | | |
| | ABS/MBS | | | | | | |
| | **Total** | −0.40% | +0.05% | −0.35% | +0.15% | −0.45% | −0.65% |
| **TOTAL** | | **−0.65%** | **+0.50%** | **−0.15%** | **+0.90%** | **−0.30%** | **+0.45%** |

Based on the results provided in this table, it would be *most appropriate* to conclude that:

**A)** the portfolio outperformed the benchmark portfolio by 45 basis points.

**B)** the yield curve likely flattened, because long- and short-duration positions underperformed the benchmark.

**C)** there was a 90 basis point gain from underweighting ABS/MBS versus corporate bonds.

**Explanation**

The total outperformance of the manager's portfolio sums to 0.45% relative to the benchmark portfolio. Interest rates likely increased more than anticipated based on the negative returns relative to duration positioning; however, we cannot determine whether the curve flattened or steepened from this data only. Similarly, we cannot determine whether the 90 basis point gain from asset allocation was attributable to overweighting or underweighting, as this information is not included in the data provided.

(Module 13.2, LOS 13.f)

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## Question #52 of 54

Question ID: 1674903

Which of the following is *least likely* to represent a benefit of carrying out performance attribution?

**A)** Additional information for passive portfolios. ✓

**B)** Quality control of active portfolios. ✗

**C)** Justification of underperformance during negative market periods. ✗

**Explanation**

After a 10-year bull market in the S&P, performance attribution is likely to become even more important as portfolio managers justify their performance and fees during periods of negative returns.

Additionally, one can consider performance attribution as a form of quality control that investors can use when evaluating active managers to see if their compensation is justified.

(Module 13.1, LOS 13.c)

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## Question #53 of 54

Question ID: 1674912

Within yield curve decomposition analysis, which of the following statements regarding the full-repricing method is *least accurate*? The full-repricing method:

**A)** requires less data than the duration-based approach. ✓

**B)** is most accurate when based on spot rates. ✗

**C)** is most accurate when based on zero-coupon curves. ✗

**Explanation**

The use of spot rates (zero-coupon curves) is known as the full-repricing method and is the most accurate measure of price changes in securities. However, it requires more data than the duration-based approach.

(Module 13.2, LOS 13.f)

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## Question #54 of 54

Question ID: 1674930

Which of the following *best* characterizes manager universes as a benchmark? Manager universes are:

**A)** a valid benchmark because they are measurable. ✗

**B)** not a valid benchmark because they are not investable. ✓

**C)** not a valid benchmark because they are not measurable. ✗

**Explanation**

Manager universes are not a valid benchmark because they are not investable, are not specified in advance, and are not unambiguous. It is also impossible to determine if they are appropriate due to the ambiguity of the median manager. Furthermore, the performance records of poor managers are dropped from manager universes, so there is an upward bias (i.e., survivorship bias) where the median manager's return is inflated. The only property of a valid benchmark that manager universes fulfill is that they are measurable.

(Module 13.3, LOS 13.j)